Last Name: First Name:

Matriculation Number:

Exam
Artificial Intelligence 2

October 2, 2026

Please ignore the QR codes; do not write on them, they are for grading support

	To be used for grading, do not write here										
prob.	1.1	1.2	2.1	2.2	3.1	4.1	4.2	5.1	5.2	Sum	grade
total	8	9	9	10	9	12	10	13	10	90	
reached											



Organizational Information

Please read the following directions carefully and acknowledge them with your signature.

1. Please place your student ID card and a photo ID on the table for checking.
2. You can reach 90 points if you fully solve all problems. You will only need 85 points for a perfect score, i.e. 5 points are bonus points.
3. No resources or tools are allowed except for a pen.
4. In particular, you are not allowed any electronic devices – phones, smart watches, earbuds, smart rings, ...on your person. Put them into your backpack or your pockets.
5. You have 90 min (sharp) for the exam.
6. Write the solutions directly on the sheets, no other paper will be graded.
7. If you have to abort the exam for health reasons, your inability to sit the exam must be certified by an examination at the University Hospital. Please notify the exam proctors and have them give you the respective form.
8. Please make sure that your copy of the exam is complete (16 pages excluding cover sheet and organizational information pages) and has a clear print. **Do not forget to add your personal information on the cover sheet and to sign this declaration.**

Declaration: With my signature I certify having received the full exam document and having read the organizational information above.

Erlangen, October 2, 2026 .....
(signature)



Organisatorisches

Bitte lesen Sie die folgenden Anweisungen genau und bestätigen Sie diese mit Ihrer Unterschrift.

1. Bitte legen Sie Ihren Studierendenausweis und einen Lichtbildausweis zur Personenkontrolle bereit!
2. Sie können 90 Punkte erreichen, wenn Sie alle Aufgaben vollständig lösen. Allerdings zählen 85 Punkte bereits als volle Punktzahl, d.h. 5 Punkte sind Bonuspunkte.
3. Es sind keine Hilfsmittel erlaubt außer einem Stift.
4. Insbesondere sind alle electronischen Geräte verboten: Telefone, smart watches, Kopfhörer, smart rings, ... Deponieren Sie diese in Ihrem Rucksack oder ihren Taschen.
5. Die Bearbeitungszeit beträgt genau 90 min.
6. Schreiben Sie die Lösungen direkt auf die ausgeteilten Aufgabenblätter. Andere Blätter werden nicht bewertet.
7. Wenn Sie die Prüfung aus gesundheitlichen Gründen abbrechen müssen, so muss Ihre Prüfungsunfähigkeit durch eine Untersuchung in der Universitätsklinik nachgewiesen werden. Melden Sie sich in jedem Fall bei der Aufsicht und lassen Sie sich das entsprechende Formular aushändigen.
8. Überprüfen Sie Ihr Exemplar der Klausur auf Vollständigkeit (16 Seiten exklusive Deckblatt und Hinweise) und einwandfreies Druckbild! **Vergessen Sie nicht, auf dem Deckblatt die Angaben zur Person einzutragen und diese Erklärung zu unterschreiben!**

Erklärung: Durch meine Unterschrift bestätige ich den Empfang der vollständigen Klausurunterlagen und die Kenntnisnahme der obigen Informationen.

Erlangen, October 2, 2026 .....
(Unterschrift)

Please consider the following guidelines to avoid losing points:

- If you continue an answer on another page, clearly give the problem number on the new page and a page reference on the old page.
- You can always ask for the translation or explanation of a non-technical word.
- If you do not want something to be graded, clearly cross it out. Adding a wrong statement to a correct solution may lead to deductions.
- The instructions "Give X", "List X" or similar mean that only X is needed. If you additionally justify your answer, we may or may not give you partial credit.
- The instruction "Assume X" means that X is information that your answer may use.
- The instruction "Model X as a Y" means that you have to describe X formally and exactly as an instance of Y using the definition of Y from the lecture.
- If you are uncertain how long or complex an answer should be, use the number of points as an indication: 1 point roughly corresponds to 1 minute.
- Within a problem, later questions can usually be answered even if you skipped earlier ones.
- In all calculation questions, you have to simplify as much as reasonably possible without a calculator. For example, $\log 2$ or 3^7 should not be calculated, but $0.4 \cdot 0.3 \cdot 0.5 = 0.06$ should be.

1 Probabilities

Problem 1.1 (Python)
Consider the Python program below where J holds the joint probability distribution of three random variables such that $J[x][y][z] = P(X = x, Y = y, Z = z)$.



1. Which probability-related operation does the function `foo` compute?

4 Points

```
def foo(J):
   res = 0
   for x in range(len(J)):
     for y in range(len(J[x])):
       for z in range(len(J[x][y])):
         if x+y>z:
           res += J[x][y][z]
   return res
```



2. Assume J is as above. Complete the Python program below to compute the conditional probability $P(X = Y \mid Z = z)$ for arbitrary z. (You may assume the value is defined.)

4 Points

```
def bar(J,z):
```

Problem 1.2 (Working with Distributions)
Consider the following joint probability distribution of random variables X, Y given in terms of a, b, c, d.

$P(X = x, Y = y)$	x	y
a	0	0
b	0	1
c	1	0
d	1	1
0	2	0
0	2	1



1. Give all subsets of the probabilities $\{a, \ldots, d\}$ that must sum to 1.

1 Points



2. In terms of $a, \ldots, d$, give $P(X = Y)$.

1 Points



3. In terms of $a, \ldots, d$, give $P(X = 0 \mid X = Y)$.

2 Points

continued on next page



4. What issue occurs when you try to compute $P(Y = 1 \mid X = 2)$?

2 Points



5. Now assume that $a \cdot d = b \cdot c$. Show that X and Y are independent.
   You must state exactly which 6 conditions you have to check. But then you only have to do the actual check for the case $x = y = 0$.

3 Points

2 Bayesian Reasoning and Utilities

Problem 2.1 (Decisions and Utility)
Consider a situation with 3 possible outcomes A, B, C. You have rational preferences that include the following:

- $C \succ B$ and $B \succ A$
- $[0.7, A\ ; 0.3, C] \sim B$

You want to construct a utility function $U(-)$ for these preferences as stipulated by Ramsey's theorem.



1. What does the property of orderability require?

2 Points



2. Assuming $U(A) = 10$, what possible values can you choose for $U(C)$?

1 Points



3. Assuming $U(A) = 10$ and $U(C) = 100$, what values can you choose for $U(B)$?

2 Points

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Now assume $U(A) = 10$, $U(B) = 13$, and $U(C) = 20$ and that $U(X)$ is the amount of money you earn in scenario X.
You estimate the three scenarios A, B, C to be equally probable.
You have the option to pay 5 money for insurance that yields a return of 10 if scenario A happens.



4. Calculate the expected utilities to decide if you should buy the insurance.

2 Points



5. You have an offer to buy the information whether scenario A happens (and if it does, you would buy the insurance).
   Calculate the value of that information.

2 Points

Problem 2.2 (Bayesian Networks)
Consider the following situation about an alarm clock:

a) It fails to ring if its batteries are empty.

b) It fails to ring if it the volume is muted.

c) You might oversleep if your alarm clock fails.

d) You might oversleep if you stay up late.

You want to model this situation as a Bayesian network using Boolean random variables.



1. Give an appropriate set of random variables and their meaning and draw the Bayesian network (using an appropriate variable order). 3 Points



2. Give the probability of the clock failing in terms of the entries of the conditional probability tables of your network. 2 Points

Now you decide to model the failing of the clock as a deterministic node.



3. Explain (in about 3 sentences) whether that decision is justified by the description, and how it changes the conditional probability tables of your model. 3 Points



4. State the probability of the clock failing in terms of the entries of the conditional probability tables of your network (with the deterministic node for the failing clock). 2 Points

3 Markovian Reasoning

Problem 3.1 (Markovian Reasoning)
Consider the following agent in a stochastic environment: The agent moves in a 3×3 grid with coordinates $0, 1, 2$ in each direction. Its possible actions are the same in each state, namely `up`, `down`, `right`, `left`. Actions that would cross the outer border of the grid do not cause any movement. Other actions succeed with a probability of 0.7 and fail without movement otherwise.
The agent starts at $\langle 0, 0 \rangle$. Its goal is to reach $\langle 2, 2 \rangle$. We use a reward of 10 for the goal and -2 otherwise, and a discount factor of $\gamma = 0.5$.



1. State the Bellman Optimality Equation for $U(s)$.

2 Points



2. Assume current utilities $U(s) = 0$ for all states s. Perform one value iteration step to calculate $U(\langle 0, 2 \rangle)$.

2 Points



3. Calculate the probability of the agent moving along to $\langle 0, 1 \rangle$ and then to $\langle 1, 1 \rangle$ by taking the actions `up` and `right`.

2 Points



4. Now assume we use the corresponding POMDP with current belief state b with $b(\langle 0, 0 \rangle) = 0.5$, $b(\langle 0, 1 \rangle) = 0.5$ and apply action `right`. Calculate the new belief state b'.

3 Points

4 Learning

Problem 4.1 (Decision Trees and Lists)
Consider an unknown natural number $N \in \{1, \ldots, 10\}$. You are allowed to ask the following attributes/questions about N:

#	Question	Answers	Note
A	Is N prime?	yes, no	1 is not a prime number
B	What is N modulo 3?	$0, 1, 2$	remainder under integer division
C	Is $N > 5$?	yes, no	
D	Is $N^2 + N - 7 = 0$?	yes, no	
E	What is the result of $\sin((N + 0.5)\pi)$?	$-1, 1$	remember $\sin(0.5\pi) = 1$, $\sin(1.5\pi) = -1$, $\sin(2.5\pi) = 1, \ldots$
F	What is N modulo 4?	$0, 1, 2, 3$	



1. Give the shortest (in terms of tree depth) decision tree for identifying N that uses at most the above questions. 3 Points



2. We can see each question as a random variable. Calculate the entropies $I(A)$, $I(C)$, and $I(D)$, i.e., the number of bits obtained by asking the question. (Simplify as much as possible without introducing approximate values.) 3 Points

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Now, to give a logical formulation, we introduce 6 binary predicate symbols $Q = A, B, C, D, E, F$ such that $Q(N, a)$ represents that if N is the unknown number, then a is the answer to question Q. We want to learn whether N is a square number, which we write as a predicate $sq(N)$.

Assume you are using inductive logic programming (ILP), using the cases for $n = 1, \ldots, 10$ as training examples, and you have already learned the first clause

$$C(N, no) \wedge B(N, 1) \Rightarrow sq(N)$$



3. Which positive examples remain, for which a second clause must be learned?

2 Points

Now assume your partially learned second clause is $C(N, yes) \Rightarrow sq(N)$.



4. Which negative examples remain for which additional literals must be added to the body?

2 Points



5. Which literal would be optimal for inductive logic programming to add to the body?

2 Points

Problem 4.2 (Regression and Loss)
Assume we want to learn the squaring function $f(n) = n^2$ using the values $n = 0, 1, \ldots, 3$ as training examples. We use (univariate) linear regression.



1. Formulate this as an inductive learning problem $\langle \mathcal{H}, T \rangle$.

2 Points



2. Briefly explain (in about 1 sentence) whether there is a consistent hypothesis.

2 Points

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From now on, consider the hypothesis $h(x) = 4 \cdot x - 3$.



3. Give the empirical loss of h using the $0/1$ loss function. 2 Points

From now on, we use the loss function $L(x, y, \hat{y}) = \max(0, \hat{y} - y) + 0.5 \cdot \max(0, y - \hat{y})$ where y is the target value for x and $\hat{y}$ the value predicted by a hypothesis.



4. Calculate the loss of h for each training example. 2 Points



5. Briefly explain (in about 2 sentences, possibly giving an example) what is unusual about the choice of L above and when we might choose such a loss function. 2 Points

5 Natural Language Processing

Problem 5.1 (Language Models and POS-Tagging)
Consider the following corpus

a dog runs. the man sees the dog. the man shouts.

You want to build a 2-gram character model using random variables c_1, c_2 based on this corpus.



1. Seen as a stationary Markov process, what is the meaning of the c_i in such a 2-gram character model and what is its order? 2 Points



2. Give those entries of the transition model of the Markov process that correspond to the case $c_1 = h$. 2 Points



3. Consider the partial word *sh*. Which character would your bigram model predict to follow? 1 Points



4. Briefly explain (in about 1 sentence) why or why not the answer to the previous question would change if you used a trigram model. 1 Points

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From now on, you apply part-of-speech tagging to the same corpus, using a stationary first-order HMM with stationary sensor model. You use the tags D (determiner), N (noun), V (verb), E (end of sentence).



5. Give the domains of the state and evidence variables. 2 Points



6. Give the transition model as a matrix. For the row/column order, assume the tags are ordered as $DNVE$.
   Assume that the initial state at the start of the corpus is E so that there is one transition for every part of speech in the corpus. 3 Points



7. In the sensor model, no column has more than 1 non-zero entry. Briefly explain (in about 2 sentences) why that is the case and why that is an unrealistic simplification for an English corpus. 2 Points

Problem 5.2 (Word Embeddings)
You work with a fixed natural language L using a set $W = \{w_1, \dots, w_k\}$ of words.



1. Give the general form of an n-dimensional word embedding for L.

1 Points

From now on, let the function e be the one hot embedding of L.



2. Explain e.

2 Points

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3. Consider a neural network with the following structure
   - input layer with k inputs
   - 1 hidden layer with n nodes
   - output layer with k outputs

   We train the network using the training examples $(e(w), e(w))$ for every $w \in W$.
   Explain (in about 3 sentences) the intention behind such a setup and its relationship to word embeddings.
   Hint: It may help to think about CBOW.

3 Points



4. Both tfidf and CBOW can be used to obtain word embeddings. Compared to the word embeddings of the previous question, they need additional data to build. Explain (in about 2 sentences) what that data is and how it affects the embedding.

2 Points



5. tfidf and CBOW use context very differently. Explain the difference (in about 2 sentences).

2 Points



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